Test Cases for Authentication and Staff Accounts Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC01	Sign in, enforce staff permissions, and expire an idle session	User barista. Try password wrong, then password123. Request staff and admin access. Advance clock by 30 minutes.	Reject wrong credentials. Allow staff access only. Expire the idle session. Store hashed credentials and tokens.	Wrong password: 401. Staff login succeeded. Admin access: 403. Session expired at 30 minutes. Hashes were verified.	Pass
TC02	Deactivate, reactivate, reset, change password, and log out	Create barista with initial-pass. Deactivate, reactivate with reset-password, change to changed-password, and log out.	Deactivate and revoke the session. Only the latest password works. Wrong current password is rejected. Logout revokes the token.	Inactive and old-password login returned 401. Wrong current password returned 403. Reset and change worked, and affected sessions were revoked.	Pass
TC03	Reject case-insensitive duplicates and salt each password	Create server-one and server-two with shared-pass. Attempt to create SERVER-ONE.	Reject the duplicate username. Store different hashes for the shared password.	Duplicate rejected with 409. Two distinct stored hashes were found.	Pass
TC04	Refresh the idle deadline on activity	Sign in as barista. Read the session at minute 29 and again at minute 58.	Each successful read refreshes the 30-minute idle window.	Both reads returned an active staff session.	Pass
TC05	Protect administrator accounts from staff edits	Create admin manager. List staff and attempt to deactivate manager through the staff module.	Exclude manager from the staff list and reject the edit.	Staff list was empty. The edit returned 404.	Pass

Test Cases for Menu Catalog Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC06	Filter the menu and show sold-out status	Search COCOA in Drinks. Search missing and category 999. Mark Mocha unavailable.	Find Mocha for COCOA and no items for the other filters. Expose sold-out availability.	COCOA found Mocha. Both unmatched filters returned no items. Availability was 0.	Pass
TC07	Validate item data and archive without losing order history	Try price -3, category 999, and a javascript: image URL. Archive Drinks before and after archiving its Mocha item.	Reject invalid inputs and prevent archiving a category with an active item. Keep the original order line and allow category name reuse.	Invalid inputs returned 400. Premature archive returned 409. Menu emptied after archival, the order retained Mocha, and Drinks could be recreated.	Pass
TC08	Rename a category and reject duplicates	Rename Drinks to Espresso bar. Try to add ESPRESSO BAR.	Show the new category on Mocha and reject the duplicate name.	Mocha showed Espresso bar. Duplicate category returned 409.	Pass
TC09	Update availability without changing price	Set Mocha unavailable. Pass the string true, then the boolean true.	Reject a non-boolean flag and restore availability at the original price.	String flag returned 400. Boolean flag restored availability at INR 149.50.	Pass
TC10	Hide archived items from public reads	Archive Mocha, search mocha, and request its details.	Omit the archived item from search and reject direct lookup.	Search returned zero items. Details lookup returned 404.	Pass

Test Cases for Cart Management Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC11	Add, update, and remove cart quantities	Add two Mochas, add one more, set quantity to four, then try 100 and set quantity to zero.	Calculate INR 448.50 for three and INR 598.00 for four. Reject 100. Remove the line at zero.	Three totalled INR 448.50. Four had a subtotal of INR 598.00. Quantity 100 returned 400. Zero emptied the cart.	Pass
TC12	Handle an item sold out after being added	Add two Mochas. Mark Mocha sold out, set quantity to three, attempt checkout, then remove it.	Reject increasing the quantity or checking out. Allow removal.	Quantity update and checkout returned 409. Removal left a zero total.	Pass
TC13	Keep different carts isolated	Create one cart with one Mocha and another with three. Look up an unknown cart ID.	Return distinct IDs and separate quantities. Treat the unknown cart as empty.	Cart IDs differed. Reads returned quantities one and three. Unknown ID returned an empty cart.	Pass
TC14	Enforce the maximum quantity on increment	Put 99 Mochas in a cart and try adding one more.	Reject the increment and keep the current quantity.	Increment returned 400. The stored quantity stayed at 99.	Pass
TC15	Use current menu price before checkout	Add two Mochas at INR 149.50 each. Change the menu price to INR 200.00 and read the cart.	Recalculate the open cart using the new price.	Cart total was INR 400.00.	Pass

Test Cases for Orders and Pickup Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC16	Validate checkout, preserve price snapshots, and replay safely	Try empty cart and phone bad. Order two Mochas, change menu price, and retry. Track with a wrong phone and add to a new cart.	Reject invalid checkout and unauthorized tracking. Replay the original order at its stored price and leave the old cart consumed.	Empty cart: 409. Bad phone: 400. Wrong tracking phone: 404. Retry used the same order at INR 299.00. A new add used a new cart.	Pass
TC17	Enforce pickup stages and one notification per transition	Try Preparing before payment and Ready before Preparing. Then set Preparing, Ready twice, and Completed. Try going backward.	Reject unpaid, skipped, and backward transitions. Record each accepted stage and notification once.	Unpaid transition returned 404. Skipped and backward stages returned 409. History and inbox had no duplicate Ready entry.	Pass
TC18	Reject a replay using another phone	Place an order from a cart with phone 9876543210, then replay it using 9876543211.	Reject the replay of the consumed cart for a different phone.	Replay returned 409.	Pass
TC19	Normalize tracking input and keep the cart token private	Track an unpaid order with its lowercase code and +91 9876543210.	Find the Pending Payment order. Omit the private cart ID and show no receipt.	Tracking succeeded with Pending Payment. Cart ID was omitted and the receipt was null.	Pass
TC20	Filter paid history by status and IST date	Complete a paid order on 11 Nov IST. Query Completed for 11 Nov, then 10 Nov, then active orders.	Return the completed order for the correct date only. Exclude it from the active queue.	Completed order appeared on 11 Nov only. The 10 Nov and active queries were empty.	Pass

Test Cases for Payments Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC21	Handle pending, failure, retry, and verified success	Start twice, verify pending, simulate failure, retry with success, and repeat verification.	Reuse the pending attempt. A failed or unverified attempt stays off the staff queue. One verified success creates one receipt.	Pending start reused the ID. Failure stayed off the queue. The retry verified twice with two total attempts, one queued order, and an INR 299.00 receipt.	Pass
TC22	Preserve order state on invalid gateway response or outage	Gateway returns a success for order other, then throws provider unavailable.	Reject the mismatch and propagate the gateway failure while leaving the attempt and order pending.	Wrong order returned 502. Provider error was raised. Attempt stayed Pending and the order stayed Pending Payment.	Pass
TC23	Roll back verification when notification writing fails	Simulate success, force the notification write to throw, restore the writer, and retry verification.	Roll back payment, order, receipt, and history changes. Let a retry succeed.	Attempt stayed Pending with no receipt and only the initial history entry. Retry returned Succeeded after the writer was restored.	Pass
TC24	Scope the payment attempt to the order and phone	Verify the first order's attempt against a second order. Simulate success with phone 9876543211.	Reject both attempts to act outside the original order and phone.	Both operations returned 404.	Pass
TC25	Reject a verified result with the wrong amount	Gateway returns the correct order ID and a success amount of 1 paise for an INR 299.00 order.	Reject the amount mismatch and leave the order without a receipt.	Verification returned 502. No receipt was issued.	Pass
TC26	Prevent another charge for a paid order	Settle an order successfully, then start another payment for it.	Reject the new payment attempt.	New attempt returned 409.	Pass

Test Cases for Demo Payment Gateway Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC27	Keep a failed terminal result immutable	Set an attempt to Failed. Try Succeeded, then try the unsupported result Forged.	Keep Failed as the terminal result. Reject an unsupported result.	Stored result stayed Failed. Forged returned 400.	Pass
TC28	Generate distinct pending references	Start two attempts for order-one at 1,200 paise. Verify the first reference.	Generate different references and report the original amount and order with Pending status.	References differed. First result was Pending for order-one and 1,200 paise.	Pass
TC29	Keep a successful terminal result immutable	Start order-two for 2,400 paise, set Succeeded, then try Failed.	Keep the successful status and the original order and amount.	Verification returned Succeeded for order-two and 2,400 paise.	Pass
TC30	Reject an unknown gateway reference	Verify the reference unknown.	Report that the provider has no matching attempt.	Verification returned 502.	Pass

Test Cases for In-App Notifications Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC31	Keep the inbox empty before payment	Place a Pending Payment order and list its notifications.	Return no notifications.	Inbox was empty.	Pass
TC32	Include the exact receipt amount and reference	Record Paid with reference DEMO-123 on an INR 299.00 order.	Include the reference and formatted payment amount in the message.	Message included DEMO-123 and the amount formatted to two decimal places.	Pass
TC33	Record pickup instructions in chronological order	Record Preparing, then Ready at the fixed test time of 10 Nov 2026, 18:45 UTC.	Return both messages in insertion order with pickup instructions and the saved timestamp.	Preparing preceded Ready. The Ready message named Sketch Café and had the expected timestamp.	Pass
TC34	Prevent duplicate messages and isolate order inboxes	Create two orders. Record Ready twice for the first order and read both inboxes.	Reject the duplicate. Leave one message in the first inbox and none in the second.	Duplicate raised a unique -constraint error. Inbox counts were one and zero.	Pass

Test Cases for Reviews and Support Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC35	Enforce review rules and support both ticket types	Review before and after completion. Try rating 6 and a duplicate. Open a standalone ticket and an order-linked ticket, then resolve the linked one.	Accept one valid review after completion. Track standalone and linked requests. Limit lookup to the correct phone.	Early and duplicate reviews returned 409. Rating 6 returned 400. Standalone stayed Open and linked was Resolved. Wrong-phone lookup returned 404.	Pass
TC36	Reject bad phone numbers and blank messages	Send Help with phone bad. Send a whitespace-only message using 9876543210.	Reject both invalid support requests.	Both requests returned 400.	Pass
TC37	Protect order-linked support	Submit a ticket for an existing order using phone 9876543211 instead of the owner's phone.	Reject the ticket linked to an order owned by another phone.	Submission returned 404.	Pass
TC38	Require and retain the first reply	Resolve a ticket with a blank reply, then Open at 9., then Different reply.	Reject the blank reply, save the first valid reply, and reject a second resolution.	Blank reply returned 400. Second resolution returned 409. The stored reply was Open at 9.	Pass

Test Cases for Sales Reports Module

Test Case ID	Test Scenario	Input	Expected Output	Actual Output	Status
TC39	Count only verified sales and group by IST date	Create a failed and a paid order for two Mochas each near the UTC-to-IST date boundary. Complete the paid pickup after reporting.	Count one paid order and INR 299.00 on 11 Nov IST. Show two units sold and one handled pickup after completion.	10 Nov showed zero orders. 11 Nov showed one order, INR 299.00, and two units. Handled pickups rose from zero to one.	Pass
TC40	Return zeros for an empty sales period	Run the 11 Nov 2026 report with no sales.	Return zero orders and total, with empty daily and best-seller lists.	Both numeric totals were zero. Both lists were empty.	Pass
TC41	Aggregate renamed items under the same item ID	Sell two Mochas, rename to Signature Mocha, then sell two more.	Combine all four units under the same item and total INR 598.00.	Signature Mocha ranked with four units. Total was INR 598.00.	Pass
TC42	Reject invalid dates before aggregation	Try missing date, 2026-2-01, and 2025-02-29. Validate leap date 2024-02-29.	Reject missing, malformed, and impossible dates. Accept the valid leap day.	All three invalid dates returned 400. The leap day was accepted.	Pass